```
\begin{thm}
  $[0,1]$ is a compact subset of $\mathbb{R}$.
\end{thm}
```

Now all the theorem-like statements produced above have the *same typographical form*—name and number in **boldface** and the body of the statement in *italics*. What if you need something like

THEOREM IX.1.1 (EUCLID). *The sum of the angles of a triangle is* 180°.

Such customization is necessitated not only by the aesthetics of the author but often by the whims of the designers in publishing houses also.

IX.2. DESIGNER THEOREMS—THE AMSTHM PACKAGE

The package amsthm affords a high level of customization in formatting theorem-like statements. Let us first look at the predefined *styles* available in this package.

IX.2.1. Ready made styles

The default style (this is what you get if you do not say anything about the style) is termed `plain` and it is what we have seen so far—name and number in boldface and body in italic. Then there is the `definition` style which gives name and number in boldface and body in roman. And finally there is the `remark` style which gives number and name in italics and body in roman.

For example if you put in the preamble

```
\usepackage{amsthm}
\newtheorem{thm}{Theorem}[section]
\theoremstyle{definition}
\newtheorem{dfn}{Definition}[section]
\theoremstyle{remark}
\newtheorem{note}{Note}[section]
\theoremstyle{plain}
\newtheorem{lem}[thm]{Lemma}
```

and then type somewhere in your document

```
\begin{dfn}
A triangle is the figure formed by joining each pair
of three non collinear points by line segments.
\end{dfn}

\begin{note}
A triangle has three angles.
\end{note}

\begin{thm}
The sum of the angles of a triangle is $180^\circ$.
\end{thm}

\begin{lem}
The sum of any two sides of a triangle is greater than or equal to the third.
\end{lem}
```

then you get

Definition IX.2.1. A triangle is the figure formed by joining each pair of three non collinear points by line segments.

Note IX.2.1. A triangle has three angles. [1]note

Theorem IX.2.1. *The sum of the angles of a triangle is* 180°.

Lemma IX.2.2. *The sum of any two sides of a triangle is greater than or equal to the third.*

Note how the `\theoremstyle` command is used to switch between various styles, especially the last `\theoremstyle{plain}` command. Without it, the previous `\theoremstyle{remark}` will still be in force when `lem` is defined and so "Lemma" will be typeset in the `remark` style.

IX.2.2. Custom made theorems

Now we are ready to roll our own "theorem styles". This is done via the `\newtheoremstyle` command, which allows us to control almost all aspects of typesetting theorem like statements. this command has nine parameters and the general syntax is

```
\newtheoremstyle%
  {name}%
  {abovespace}%
  {belowspace}%
  {bodyfont}%
  {indent}%
  {headfont}%
  {headpunct}%
  {headspace}%
  {custom-head-spec}%
```

The first parameter *name* is the name of the new *style*. Note that it is *not* the name of the *environment* which is to be used later. Thus in the example above `remark` is the name of a new style for typesetting theorem like statements and `note` is the name of the environment subsequently defined to have this style (and `Note` is the name of the statement itself).

The next two parameters determine the vertical space between the theorem and the surrounding text—the *abovespace* is the space from the preceding text and the *belowspace* the space from the following text. You can specify either a rigid length (such as `12pt`) or a rubber length (such as `\baselineskip`) as a value for either of these. Leaving either of these empty sets them to the "usual values" (Technically the `\topsep`).

The fourth parameter *bodyfont* specifies the font to be used for the body of the theorem-like statement. This is to be given as a *declaration* such as `\scshape` or `\bfseries` and *not* as a *command* such as `\textsc` or `\textbf`. If this is left empty, then the main text font of the document is used.

The next four parameters refer to the *theoremhead*—the part of the theorem like statement consisting of the name, number and the optional note. The fifth parameter *indent* specifies the indentation of *theoremhead* from the left margin. If this is empty, then there is no indentation of the *theoremhead* from the left margin. The next parameter specifies the font to be used for the *theoremhead*. The comments about the parameter *bodyfont*, made in the previous paragraph holds for this also. The parameter *headpunct*

(the seventh in our list) is for specifying the *punctuation* after the theoremhead. If you do not want any, you can leave this empty. The last parameter in this category (the last but one in the entire list), namely *headspace*, determines the (horizontal) space to be left between the *theoremhead* and the *theorembody*. If you want only a normal interword space here put a *single blank space* as `{ }` in this place. (Note that it is not the same as leaving this *empty* as in `{}`.) Another option here is to put the command `\newline` here. Then instead of a space, you get a linebreak in the output; that is, the *theoremhead* will be printed in a line by itself and the *theorembody* starts from the next line.

The last parameter *custom-head-spec* is for customizing *theoremheads*. Since it needs some explanation (and since we are definitely in need of some breathing space), let us now look at a few examples using the eight parameters we've already discussed.

It is almost obvious now how the last theorem in Section 1 (see Page 111) was designed. It was generated by

```
\newtheoremstyle{mystyle}{}{}{\slshape}{}{\scshape}{.}{ }{}
\theoremstyle{mystyle}
\newtheorem{mythm}{Theorem}[section]
\begin{mythm}
 The sum of the angles of a triangle is $180^\circ$.
\end{mythm}
```

As another example, consider the following

```
\newtheoremstyle{mynewstyle}{12pt}{12pt}{\itshape}%
  {}{\sffamily}{:}{\newline}{}
\theoremstyle{mynewstyle}
\newtheorem{mynewthm}{Theorem}[section]
\begin{mynewthm}[Euclid]
 The sum of the angles of a triangle is $180^\circ$.
\end{mynewthm}
```

This produces

Theorem IX.2.1 (Euclid):
The sum of the angles of a triangle is 180°.

Do you need anything more? Perhaps yes. Note that *theoremhead* includes the optional note to the theorem also, so that the font of the number and name of the theorem-like statement and that of the optional note are always the same. What if you need something like

Cauchy's Theorem (Third Version). *If G is a simply connected open subset of $\mathbb{C}$, then for every closed rectifiable curve γ in G, we have*

$$\int_\gamma f = 0.$$

It is in such cases, that the last parameter of `\newtheoremstyle` is needed. Using it we can separately customize the name and number of the theorem-like statement and also the optional note. The basic syntax for setting this parameter is

`{`*commands*`#1`*commands*`#2`*commands*`#3}`

where `#1` corresponds to the name of the theorem-like statement, `#2` corresponds to its number and `#3` corresponds to the optional note. We are here actually supplying the replacement text for a command `\thmhead` which has three arguments. It is as if we are defining

```
\renewcommand{\thmhead}[3]{...#1...#2...#3}
```

but without actually typing the `\renewcommand{\thmhead}[3]`. For example the theorem above (Cauchy's Theorem) was produced by

```
\newtheoremstyle{nonum}{}{}{\itshape}{}{\bfseries}{.}{ }{#1 (\mdseries #3)}
\theoremstyle{nonum}
\newtheorem{Cauchy}{Cauchy's Theorem}

\begin{Cauchy}[Third Version]
If $G$ is a simply connected open subset of $\mathbb{C}$, then for every closed
rectifiable curve $\gamma$ in $G$, we have
 \begin{equation*}
   \int_\gamma f=0.
 \end{equation*}
\end{Cauchy}
```

Note that the absence of `#2` in the *custom-head-spec*, suppresses the theorem number and that the *space* after `#1` and the command `(\mdseries#3)` sets the optional note in medium size within parentheses and with a preceding space.

Now if you try to produce

> **Riemann Mapping Theorem.** *Every open simply connected proper subset of* $\mathbb{C}$ *is analytically homeomorphic to the open unit disk in* $\mathbb{C}$.

by typing

```
\theoremstyle{nonum}
\newtheorem{Riemann}{Riemann Mapping THeorem}

\begin{Riemann}Every open simply connected proper subset of $\mathbb{C}$ is analytically
homeomorphic to the open unit disk in $\mathbb{C}$.
\end{Riemann}
```

you will get

> **Riemann Mapping Theorem** (). *Every open simply connected proper subset of* $\mathbb{C}$ *is analytically homeomorphic to the open unit disk in* $\mathbb{C}$.

Do you see what is happened? In the `\theoremstyle{diffnotenonum}`, the parameter controlling the *note* part of the *theoremhead* was defined as `(\mdseries #3)` and in the `\newtheorem{Riemann}`, there is no optional note, so that in the output, you get an empty "note", *enclosed in parantheses* (and also with a preceding space).

To get around these difficulties, you can use the commands `\thmname`, `\thmnumber` and `\thmnote` *within* the {*custom-head-spec*} as

```
{\thmname{commands#1}%
\thmnumber{commands#2}%
\thmnote{commands#3}}
```

Each of these three commands will typeset its argument *if and only if the corresponding argument in the* `\thmhead` *is non empty*. Thus the correct way to get the **Riemann Mapping theorem** in Page 114 is to input

```
\newtheoremstyle{newnonum}{}{}{\itshape}{}{\bfseries}{.}{ }%
    {\thmname{#1}\thmnote{ (\mdseries #3)}}

\theoremstyle{newnonum}
\newtheorem{newRiemann}{Riemann Mapping Theorem}

\begin{newRiemann} Every open simply connected proper subset of $\mathbb{C}$ is
analytically homeomorphic to the open unit disk in $\mathbb{C}$.
\end{newRiemann}
```

Then you can also produce **Cauchy's Theorem** in Page 113 by typing

```
\theoremstyle{newnonum}
\newtheorem{newCauchy}{Cauchy's Theorem}

\begin{newCauchy}[Third Version]If $G$ is a simply connected open subset of
$\mathbb{C}$, then for every closed rectifiable curve $\gamma$ in $G$, we have
\begin{equation*}
 \int_\gamma f=0
\end{equation*}
\end{newCauchy}
```

The output will be exactly the same as that seen in Page 113. Now suppose you want to highlight certain theorems from other sources in your document, such as

> **Axiom 1 in [1].** *Things that are equal to the same thing are equal to one another.*

This can be done as follows:

```
\newtheoremstyle{citing}{}{}{\itshape}{}{\bfseries}{.}{ }{\thmnote{#3}}

\theoremstyle{citing}
\newtheorem{cit}{}

\begin{cit}[Axiom 1 in \cite{eu}]
 Things that are equal to the same thing are equal to one another.
\end{cit}
```

Of course, your *bibliography* should include the citation with *label* `eu`.

IX.2.3. There is more!

There are some more predefined features in amsthm package. In all the different examples we have seen so far, the *theorem number* comes after the *theorem name*. Some prefer to have it the other way round as in

> **IX.2.1 Theorem (Euclid).** *The sum of the angles in a triangle is* 180°.

This effect is produced by the command `\swapnumbers` as shown below:

```
\swapnumbers
\theoremstyle{plain}
\newtheorem{numfirstthm}{Theorem}[section]

\begin{numfirstthm}[Euclid]
The sum of the angles in a triangle is $180^\circ$
\end{numfirstthm}
```

Note that the `\swapnumbers` command is a sort of toggle-switch, so that once it is given, *all subsequent theorem-like statements* will have their numbers first. If you want it the other way for some other theorem, then give `\swapnumbers` again before its definition.

A quick way to suppress *theoremnumbers* is to use the `\newtheorem*` command as in

```
\newtheorem*{numlessthm}{Theorem}[section]

\begin{numlessthm}[Euclid]
The sum of the angles in a triangle is $180^\circ$.
\end{numlessthm}
```

to produce

Euclid. *The sum of the angles in a triangle is* 180°.

Note that this could also be done by leaving out `#2` in the *custom-head-spec* parameter of `\newtheoremstyle`, as seen earlier.

We have been talking only about *theorems* so far, but Mathematicians do not live by theorems alone; they need *proofs*. The amsthm package contains a predefined `proof` environment so that the proof of a theorem-like statement can be enclosed within `\begin{proof} ... \end{proof}` commands as shown below:

```
\begin{thmsec}
The number of primes is infinite.
\end{thmsec}

\begin{proof}
Let $\{p_1,p_2,\dotsc p_k\}$ be a finite set of primes. Define $n=p_1p_2\dotsm
p_k+1$. Then either $n$ itself is a prime or has a prime factor. Now  $n$ is
neither equal to nor is divisible by any of the primes $p_1,p_2,\dotsc p_k$ so
that in either case, we get a prime different from $p_1,p_2,\dotsc p_k$. Thus
no finite set of primes can include all the primes.
\end{proof}
```

to produce the following output

Theorem IX.2.3. *The number of primes is infinite.*

Proof. Let $\{p_1, p_2, \dots p_k\}$ be a finite set of primes. Define $n = p_1 p_2 \cdots p_k + 1$. Then either n itself is a prime or has a prime factor. Now n is neither equal to nor is divisible by any of the primes $p_1, p_2, \dots p_k$ so that in either case, we get a prime different from $p_1, p_2, \dots p_k$. Thus no finite set of primes can include all the primes. □

There is an optional argument to the `proof` environment which can be used to change the *proofhead*. For example,

```
\begin{proof}[\textsc{Proof\,(Euclid)}:]
\begin{proof}
Let $\{p_1,p_2,\dotsc p_k\}$ be a finite set of primes. Define $n=p_1p_2\dotsm
p_k+1$. Then either $n$ itself is a prime or has a prime factor. Now  $n$ is
neither equal to nor is divisible by any of the primes $p_1,p_2,\dotsc p_k$ so
that in either case, we get a prime different from $p_1,p_2,\dotsc p_k$. Thus
no finite set of primes can include all the primes.
\end{proof}
```

produces the following

PROOF (EUCLID): Let $\{p_1, p_2, \dots p_k\}$ be a finite set of primes. Define $n = p_1 p_2 \cdots p_k + 1$. Then either n itself is a prime or has a prime factor. Now n is neither equal to nor is divisible by any of the primes $p_1, p_2, \dots p_k$ so that in either case, we get a prime different from $p_1, p_2 \dots p_k$. Thus no finite set of primes can include all the primes. □

Note that the end of a proof is *automatically* marked with a □ which is defined in the package by the command `\qedsymbol`. If you wish to change it, use `\renewcommand` to redefine the `\qedsymbol`. Thus if you like the original "Halmos symbol" ∎ to mark the ends of your proofs, include

```
\newcommand{\halmos}{\rule{1mm}{2.5mm}}
\renewcommand{\qedsymbol}{\halmos}
```

in the preamble to your document.

Again, the placement of the `\qedsymbol` at the *end* of the last line of the proof is done via the command `\qed`. The default placement may not be very pleasing in some cases as in

Theorem IX.2.4. *The square of the sum of two numbers is equal to the sum of their squares and twice their product.*

Proof. This follows easily from the equation

$$(x + y)^2 = x^2 + y^2 + 2xy$$

□

It would be better if this is typeset as

Theorem IX.2.5. *The square of the sum of two numbers is equal to the sum of their squares and twice their product.*

Proof. This follows easily from the equation

$$(x + y)^2 = x^2 + y^2 + 2xy$$

□

which is achieved by the input shown below:

```
\begin{proof}
This follows easily from the equation
\begin{equation}
(x+y)^2=x^2+y^2+2xy\tag*{\qed}
\end{equation}
\renewcommand{\qed}{}
\end{proof}
```

For this trick to work, you must have loaded the package `amsmath` *without* the `leqno` option. Or, if you prefer

Proof. This follows easily from the equation

$$(x + y)^2 = x^2 + y^2 + 2xy \quad \square$$

Then you can use